

Emergence of connectedness in Social Networks

Prerequisite: Basics of NetworkX

The emergence of connectedness is to check whether the graph is connected or not. It says that in a graph of N nodes, we just need $N \log N$ edges to make graph connected.

Approach:

The following algorithm to be followed:

1. Take any random graph with N number of nodes.
Now choose any node as a starting node.
3. Now add random edges to the graph and check every time if the graph is connected or not.
4. If the graph is not connected then repeat steps 2 and 3.
5. If the graph is connected then check the number of edges that are added and this will be equal to $N \log N$.

Now check both the plots will be almost similar.

Code for Checking connectedness: